discussion session: 3/4/24 textbook: 7.7 Approximate integration

ex 1: $\int_{-2}^{6} \left(\frac{1}{2}x^2 + x^3 - 1\right) dx$, $n=4$

$\Delta x = \frac{b-a}{n} = \frac{6-(-2)}{4} = 2$

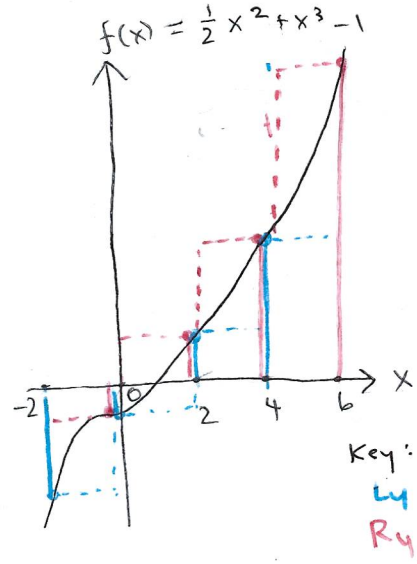
Δx

$x_0=-2 \quad x_1=0 \quad x_2=2 \quad x_3=4 \quad x_4=6$

$$L_4 = \Delta x \sum_{i=1}^{4} f(x_{i-1}) = 2\left(f(-2)+f(0)+f(2)+f(4)\right)$$
$$= 2\left[(2-8-1)+(-1)+(2+8-1)+(8+64-1)\right]$$
$$= 2(-7-1+9+71)$$
$$= 2(72)$$
$$= \boxed{144}$$

$$R_4 = \Delta x \sum_{i=1}^{4} f(x_i) = 2\left(f(0)+f(2)+f(4)+f(6)\right)$$
$$= 2\left[(-1)+9+71+(18+216-1)\right]$$
$$= 2(-1+9+71+233)$$
$$= 2(312)$$
$$= \boxed{624}$$

(actual value of $\int_{-2}^{6} f(x)\,dx = 349\frac{1}{3}$)



ex 2: $\int_{-1}^{4} x - \frac{1}{x+2}\,dx$, $n=5$

$\Delta x = \frac{b-a}{n} = \frac{4-(-1)}{5} = 1$

$-\frac{1}{2} \quad \frac{1}{2} \quad \frac{3}{2} \quad \frac{5}{2} \quad \frac{7}{2}$

Note $\bar{x}_i = \frac{1}{2}(x_{i-1}+x_i)$

$$M_5 = \Delta x \sum_{i=1}^{4} f(\bar{x}_i) = 1\left(f(-\tfrac{1}{2})+f(\tfrac{1}{2})+f(\tfrac{3}{2})+f(\tfrac{5}{2})+f(\tfrac{7}{2})\right)$$
$$= \left(-\frac{1}{2} - \frac{2}{3} + \frac{1}{2} - \frac{2}{5} + \frac{3}{2} - \frac{2}{7} + \frac{5}{2} - \frac{2}{9} + \frac{7}{2} - \frac{2}{11}\right.$$
$$= \frac{15}{2} - \frac{2}{3} - \frac{2}{5} - \frac{2}{7} - \frac{2}{9} - \frac{2}{11}$$
$$= 5\frac{5153}{6930}$$
$$\approx 5.7436$$

(actual value: ≈ 5.708)

error bound:

Find K:

$f(x) = x - (x+2)^{-1}$

$f'(x) = 1-(x+2)^{-2}$

$f''(x) = -2(x+2)^{-3}$

On $[-1,4]$, $|f''(x)| = \left|\frac{2}{(x+2)^3}\right| \le \frac{2}{(-1+2)^3} = \frac{2}{1^3} = 2$

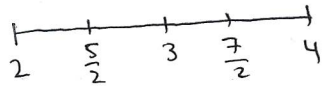
minimize denominator to get max. $|f''(x)|$ for $x \in [-1,4]$

So, $K = 2$

$$|E_M| \le \frac{K(b-a)^3}{24n^2}$$
$$= 2 \cdot \frac{(4-(-1))^3}{24 \cdot 5^2}$$
$$= 2 \cdot \frac{5^3}{24 \cdot 5^2}$$
$$= \frac{5}{12}$$
$$\approx 0.417$$

ex 3: $\int_2^4 \frac{1}{2}x^2 + 2x^{\frac{1}{2}}\,dx$, $n = 4$

$\Delta x = \frac{b-a}{n} = \frac{4-2}{4} = \frac{1}{2}$



$$T_4 = \frac{\Delta x}{2}\left[f(2) + 2f(\tfrac{5}{2}) + 2f(3) + 2f(\tfrac{7}{2}) + f(4)\right]$$

$$= \frac{1}{2}\cdot\frac{1}{2}\left(\frac{1}{2}\cdot 2^2 + 2\sqrt{2} + \left(\frac{5}{2}\right)^2 + 4\sqrt{\frac{5}{2}} + 9 + 4\sqrt{3} + \left(\frac{7}{2}\right)^2 + 4\sqrt{\frac{7}{2}} + \frac{1}{2}\cdot 4^2 + 2\sqrt{4}\right)$$

$$= \frac{1}{4}\left(2 + 2\sqrt{2} + \frac{25}{4} + 4\sqrt{\frac{5}{2}} + 9 + 4\sqrt{3} + \frac{49}{4} + 4\sqrt{\frac{7}{2}} + 8 + 4\right)$$

$$= \frac{1}{4}\left(23 + \frac{74}{4} + 2\sqrt{2} + 4\sqrt{\frac{5}{2}} + 4\sqrt{3} + 4\sqrt{\frac{7}{2}}\right)$$

≈ 16.2661 actual value: ≈ 16.2288

error bound:

$f(x) = \frac{1}{2}x^2 + 2x^{\frac{1}{2}}$

$f'(x) = x + x^{-\frac{1}{2}}$

$f''(x) = 1 - \frac{1}{2}x^{-\frac{3}{2}}$

On $[2,4]$, $|f''(x)| = \left|1 - \frac{1}{2x^{3/2}}\right| \le \left|1 - \frac{1}{2(4)^{3/2}}\right| = \left|1 - \frac{1}{2}\cdot\frac{1}{8}\right| = 1 - \frac{1}{16} = \frac{15}{16} = K$

$$|E_T| \le \frac{K(b-a)^3}{12n^2}$$

$$= \frac{15}{16}\cdot\frac{(4-2)^3}{12\cdot 4^2}$$

$$= \frac{15}{16}\cdot\frac{2^3}{12\cdot 4^2} \approx 0.0391$$

For same integral,

$$S_4 = \frac{\Delta x}{3}\left[f(2) + 4f(\tfrac{5}{2}) + 2f(3) + 4f(\tfrac{7}{2}) + f(4)\right]$$

$$= \frac{1}{3}\cdot\frac{1}{2}\left[2 + 2\sqrt{2} + 2\left(\frac{25}{4} + 4\sqrt{\frac{5}{2}}\right) + 9 + 4\sqrt{3} + 2\left(\frac{49}{4} + 4\sqrt{\frac{7}{2}}\right) + 12\right]$$

$$= \frac{1}{6}\left(23 + 2\sqrt{2} + \frac{25}{2} + 8\sqrt{\frac{5}{2}} + 4\sqrt{3} + \frac{49}{2} + 8\sqrt{\frac{7}{2}}\right)$$

$$= \frac{1}{6}\left(60 + 2\sqrt{2} + 8\sqrt{\frac{5}{2}} + 4\sqrt{3} + 8\sqrt{\frac{7}{2}}\right)$$

≈ 16.2287

We already found

$f''(x) = 1 - \frac{1}{2}x^{-3/2}$.

$f'''(x) = \frac{3}{4}x^{-5/2}$

$f^{iv}(x) = \frac{15}{8}x^{-7/2}$

On $[2,4]$, $|f^{iv}(x)| = \left|\frac{15}{8x^{7/2}}\right| = \frac{15}{8x^{7/2}} \le \frac{15}{8(2)^{7/2}} = \frac{15}{8\cdot 2^3\sqrt{2}} = \frac{15}{64\sqrt{2}}$

To maximize this fraction, minimize the denominator.

So, $|E_S| \le \frac{K(b-a)^5}{180n^4} = \frac{15}{64\sqrt{2}}\cdot\frac{2^5}{180\cdot 4^4} = \frac{15\cdot 2^5}{2^6\sqrt{2}\cdot 180\cdot 4^4} = \frac{1}{2\sqrt{2}\cdot 12\cdot 4^4} = \frac{1}{6144\sqrt{2}} \approx 0.000115$